

CertiCut: Capacitated K-Way Independent-QPD Circuit Cutting with Anytime Overhead Bounds

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Abstract—Circuit cutting trades quantum-device width for a quasiprobability sampling overhead that can grow multiplicatively with severed interactions. We formulate static capacitated K -way logical-qubit partitioning under a fixed independent per-gate quasiprobability-decomposition (QPD) model. Aggregating gate occurrences by interaction pair in the log domain yields a weighted capacitated graph-partitioning MIP. For any $K \geq 2$, feasible capacity bounds, and valid global bounds $LB \leq OPT \leq UB$, an incumbent satisfies $\Gamma_{inc}/\Gamma^* \leq \exp(UB - LB)$. Controlled and algorithm-derived instances demonstrate that minimizing cut count can select partitions with substantially larger modeled placement regret (up to a factor of 1.04×10^6). On a 144-run heterogeneous matrix through $n = 60$ and $K = 5$, 101/144 runs reach solver-tolerance closure within 10 s; open cases retain explicit finite factors. Furthermore, semantically equivalent circuit representations can alter the optimal fragment placement itself. We establish an upper bound on representation regret, prove that a perturbation $\|\delta\|_1$ smaller than the primary optimality margin m_a guarantees optimal-partition containment ($\arg \min J_b \subseteq \arg \min J_a$), and prove a positive-scaling invariance corollary explaining QAOA placement stability ($R = 1$) despite a $2 \times$ cost shift. A tie-safe two-stage lexicographic MIP confirms placement reversals on quantum phase estimation ($R = 729 \times$) and Grover circuits ($\log_{10} R = 62.8$), while ruling out single-solve tie-breaking false positives on QFT circuits.

Index Terms—quantum circuit cutting, quasiprobability decomposition, capacitated graph partitioning, anytime optimization, mixed-integer programming

I. INTRODUCTION

Near-term quantum processors remain limited in qubit count, fidelity, and connectivity [1]. Circuit cutting mitigates the width constraint by decomposing a circuit into smaller, independently executable fragments and recovering the target quantity through classical post-processing [2], [3], [4]. This tradeoff can be expensive: every severed interaction contributes a gate-dependent quasiprobability sampling factor, and these factors multiply across cuts. Minimizing the number of cuts therefore need not minimize the resulting sampling overhead.

Automatic cutting workflows commonly report a partition and a solver-specific status. Although generic mixed-integer programming (MIP) solvers expose primal and dual bounds, their implications for circuit-cutting resource overhead are not explicit in the solver output. Such a translation matters because an additive objective gap can represent orders of magnitude in multiplicative sampling cost. Working in the log-overhead domain makes the connection exact: a global optimization gap immediately yields a worst-case overhead ratio.

We present CERTICUT, a solver-independent resource-semantics framework and a capacitated K -way independent-QPD MIP. The formulation targets logical interaction graphs, supports arbitrary declared fragment counts and per-fragment lower/upper capacities, and uses independent gate-level QPD costs from Qiskit Addon Cutting 0.10 [5]. In exact arithmetic, global bounds at a safe stopping boundary imply $F = \exp(UB - LB)$; runtime outputs are floating-point solver-tolerance quantities. The production optimizer is SCIP. A specialized B2S branch-and-bound implementation remains a $K = 2$ balanced-bisection reference for polyhedral ablation, not a production-performance claim.

This work makes three contributions:

- 1) **Part I: Capacitated K -way formulation and model-overhead semantics.** For arbitrary $K \geq 2$, lower/upper fragment capacities, fixed representation, and independent per-gate QPD, we give an exact assignment/locality MIP. In exact arithmetic, a global gap gives $\Gamma_{inc}/\Gamma^* \leq \exp(UB - LB)$; implementation outputs are labelled solver-tolerance bounds.
- 2) **Part II: Representation-induced placement regret and stability theorem.** We define tie-safe cross-representation placement regret $R_{a \rightarrow b}$ and prove that an interaction-weight perturbation $\|\delta\|_1 < m_a$ guarantees placement stability ($\arg \min J_b \subseteq \arg \min J_a$). A positive-scaling corollary explains QAOA stability. Using a two-stage lexicographic MIP across $\tau_{opt} \in \{10^{-8}, 10^{-9}, 10^{-10}\}$, we show confirmed placement reversals on QPE and Grover while ruling out single-solve tie-breaking artifacts on QFT.
- 3) **Empirical characterization and backend boundary.** We establish count-versus-QPD reversals on controlled and algorithm-derived circuits, compare the specialized reference with mature SCIP, evaluate capacitated K -way scaling through $n = 60$ and $K = 5$, and quantify the regret of a fast capacity-exact heuristic using global lower bounds.

The remainder develops the formulation and bound semantics, then evaluates objective reversals and K -way backend behavior. Throughout, we separate circuit-cutting model semantics from optimization-backend performance.

CERTICUT Conceptual Model

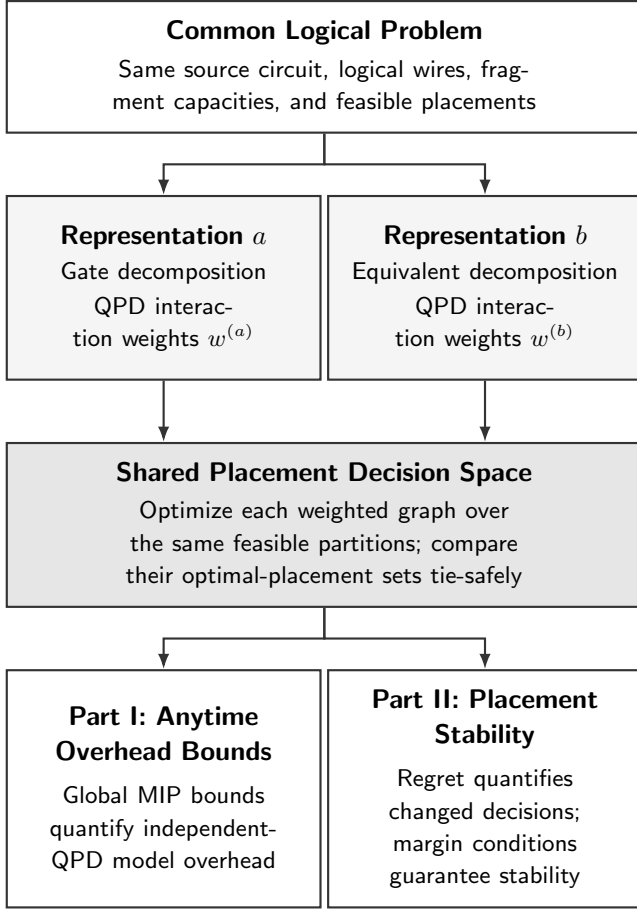


Fig. 1. CERTICUT conceptual model. A common logical circuit and feasible placement space induce representation-specific independent-QPD interaction weights. Part I optimizes each fixed-representation instance and interprets valid global bounds through $F = \exp(\text{UB} - \text{LB})$. Part II compares the resulting optimal-placement sets tie-safely to quantify representation-induced regret or establish placement stability.

II. BACKGROUND AND PROBLEM FORMULATION

A. Independent QPD Circuit Cutting

Gate cutting replaces selected two-qubit channels by quasiprobability decompositions (QPDs) over fragment-local channels [2], [6]. For a cuttable channel $\mathcal{U}_g = \sum_i a_{g,i} \mathcal{E}_{g,i}$, the corresponding independent sampling overhead is

$$\rho_g = \left(\sum_i |a_{g,i}| \right)^2. \quad (1)$$

For a partition P with crossed-gate set $S(P)$, independent sampling across gates gives

$$\Gamma(P) = \prod_{g \in S(P)} \rho_g, \quad J(P) = \log \Gamma(P) = \sum_{g \in S(P)} \log \rho_g. \quad (2)$$

This objective matches the independent gate-level model in Qiskit Addon Cutting 0.10 [5]; it is not a model of joint decompositions, finite-shot variance, or hardware noise. Because the QPD coefficient one-norm is at least one, $\rho_g \geq 1$ and all log weights are nonnegative. We validated the gate costs against the Qiskit QPD registry: $\rho_{\text{CX}} = \rho_{\text{CZ}} = 9$, $\rho_{\text{iSWAP}} = \rho_{\text{DCX}} = 49$, $\rho_{\text{CS}} = 3 + 2\sqrt{2} \approx 5.828$, and $\rho_{\text{RZZ}(\theta)} = (1 + 2|\sin \theta|)^2$ for numerically bound rotations.

B. Log-Domain Interaction Graph

Let $V = \{0, \dots, n-1\}$ be the circuit qubits, and let G_{ij} contain the two-qubit gate occurrences acting on pair (i, j) . We assign the nonnegative interaction weight

$$w_{ij} = \sum_{g \in G_{ij}} \log \rho_g. \quad (3)$$

For any fragment map $f : V \rightarrow \{1, \dots, K\}$, the objective becomes

$$J(P) = \sum_{i < j} w_{ij} \cdot \mathbb{I}(f(i) \neq f(j)). \quad (4)$$

The binary bisection identity $\mathbb{I}(f(i) \neq f(j)) = |z_i - z_j|$ is recovered when $K = 2$.

Proposition 1 (Objective equivalence). *Under independent gate-level QPD, minimizing (2) is equivalent to minimizing (4).*

Proof. Every $g \in G_{ij}$ is cut exactly when $f(i) \neq f(j)$. Regrouping the gate-level sum by qubit pair yields (4). $\square$

C. Capacitated K -Way Partition Problem

Let $K \geq 2$ be the declared fragment count. Binary assignment $a_{ik} = 1$ denotes that qubit i belongs to fragment k , with capacities $L_k \leq \sum_i a_{ik} \leq U_k$. For each aggregated interaction edge $e = (i, j)$, locality variable $y_{ek} = 1$ denotes that both endpoints occupy fragment k , while $z_e = 1$ denotes a crossed interaction. The exact independent-QPD MIP is

$$\begin{aligned} \min \quad & \sum_{e \in E} w_e z_e \\ \text{s.t.} \quad & \sum_{k=1}^K a_{ik} = 1 \quad \forall i, \\ & L_k \leq \sum_i a_{ik} \leq U_k \quad \forall k, \\ & y_{ek} \leq a_{ik}, \quad y_{ek} \leq a_{jk} \quad \forall e = (i, j), k, \\ & y_{ek} \geq a_{ik} + a_{jk} - 1 \quad \forall e = (i, j), k, \\ & z_e + \sum_{k=1}^K y_{ek} = 1 \quad \forall e, \\ & a_{ik}, y_{ek}, z_e \in \{0, 1\}. \end{aligned} \quad (5)$$

The three inequalities are the exact binary AND hull. Since each endpoint has exactly one assignment, exactly one y_{ek} is one iff both endpoints share a fragment; otherwise all locality variables are zero and $z_e = 1$. Thus (5) exactly optimizes (4) for arbitrary feasible capacities.

Proposition 2 (Objective equivalence for capacitated K -way partitioning). *For any $K \geq 2$ and feasible capacities,*

the objective of (5) equals the independent gate-level log overhead (2).

Proof. The assignment/locality constraints set $z_e = 1$ exactly when endpoints of aggregate edge e are in different fragments. Equation (3) sums the log overhead of every gate occurrence in G_e . Regrouping (2) by edge yields $\sum_e w_e z_e$. $\square$

Problem (5) is NP-hard because exact-balanced weighted bisection is its $K = 2$ special case.

D. Fixed-Capacity Cardinality and the Balanced $K = 2$ Special Case

If fragment sizes are fixed exactly as $|F_k| = n_k$, every feasible partition crosses

$$\sum_{i < j} \mathbb{I}(f(i) \neq f(j)) = \binom{n}{2} - \sum_{k=1}^K \binom{n_k}{2} = \frac{1}{2} \left(n^2 - \sum_{k=1}^K n_k^2 \right) \quad (6)$$

complete-graph pairs. This is a structural identity, not a constraint for lower/upper-only capacities. Exact-balanced bisection is the special case $K = 2$, $n_1 = n_2 = n/2$, yielding $n^2/4$.

Proposition 3 (Trivial sparse root relaxation). *For exact capacities $L_k = U_k = n_k$, the LP relaxation of the sparse edge-only formulation (5) has objective zero.*

Proof. Set $a_{ik} = n_k/n$ and $y_{ek} = n_k/n$ for every qubit i , edge e , and fragment k , then set $z_e = 0$. Assignment and capacity equalities hold since $\sum_k n_k = n$. The upper AND inequalities hold at equality; the lower inequality holds because $n_k/n \geq 2n_k/n - 1$. Finally, $z_e + \sum_k y_{ek} = 1$. Thus this is LP feasible with zero nonnegative objective. $\square$

We therefore tested an all-pair extension of (5) that imposes (6) with auxiliary nonedge variables, complete-metric triangles, and $a_{0,1} = 1$ for equal capacities. The extension is valid but dense. On a 36-record $n \in \{20, 32\}$ ablation at 5 s, the base formulation closes 31 records, symmetry-only closes 29, and the cardinality or cardinality-plus-metric variants close 6 each. Thus production E9 uses SCIP's sparse assignment/locality formulation with the valid equal-capacity anchor only; the anchor removes only a subset of label symmetry, and stronger symmetry handling remains future work because it showed no end-to-end gain in this ablation.

E. Exact Balanced $K=2$ Reference Problem

The specialized reference implementation uses the $K = 2$ exact-balanced special case. All legacy reference circuits have even n ; it requires $n/2$ qubits per fragment and fixes $z_0 = 0$ to remove label symmetry:

$$\min_{z \in \{0,1\}^n} \sum_{i < j} w_{ij} |z_i - z_j| \quad \text{s.t.} \quad \sum_i z_i = n/2, \quad z_0 = 0. \quad (7)$$

Every feasible bisection cuts exactly $(n/2)^2$ pairs in the complete-pair representation:

$$\sum_{i < j} x_{ij} = n^2/4. \quad (8)$$

This identity is central to the strengthened relaxation below.

III. OPTIMIZATION FORMULATIONS AND BOUND SEMANTICS

A. Exact MILP Validation Oracle

We use a compact MIP and brute-force enumeration as independent validation oracles. For one pair (i, j) , the convex hull of $x_{ij} = |z_i - z_j|$ is described by

$$\begin{aligned} x_{ij} &\geq z_i - z_j, & x_{ij} &\geq z_j - z_i, \\ x_{ij} &\leq z_i + z_j, & x_{ij} &\leq 2 - z_i - z_j. \end{aligned} \quad (9)$$

The implementation uses an equivalent one-hot assignment model. Its MIP retains only the two lower inequalities in (9) and declares x_{ij} binary; for positive-weight interacting pairs, minimization sets x_{ij} to its exact XOR value. Zero-weight pairs do not affect the objective. SciPy/HIGHS solves this MIP, while exhaustive enumeration supplies a second oracle through 16 qubits. Neither oracle contributes bounds to the anytime results.

B. Basic Assignment Relaxation

The basic reference LP, denoted A0, uses one-hot assignment variables on qubits and cut variables only for interacting pairs. It retains exact balance and the two lower XOR inequalities from (9), then relaxes all variables to $[0, 1]$. Because the weights are nonnegative, this is a valid lower relaxation. It is nevertheless weak: fractional assignments can make many weighted cut variables small simultaneously.

C. B2S Polyhedral Strengthening

B2S lifts A0 to the complete pair space, imposes the full XOR hull (9), and adds two families of valid constraints. To isolate the added families independently of this representation change, Sec. V compares them against a complete-pair baseline, denoted CP0.

Balanced-cut cardinality. Equality (8) fixes the total cut mass over all qubit pairs.

Metric inequalities. For each triple (i, j, k) , the cut polytope satisfies [7]

$$x_{ij} - x_{ik} - x_{jk} \leq 0, \quad (10)$$

$$x_{ik} - x_{ij} - x_{jk} \leq 0, \quad (11)$$

$$x_{jk} - x_{ij} - x_{ik} \leq 0, \quad (12)$$

$$x_{ij} + x_{ik} + x_{jk} \leq 2. \quad (13)$$

Among three binary labels, the number of pairwise disagreements is zero or two; inequalities (10)–(13) exclude the fractional analogues of one or three disagreements. B2S solves the root LP, separates violated inequalities, and repeats until no violation remains or the separation budget is exhausted. The resulting pool is then reused throughout the search tree; no node-level separation is performed.

Proposition 4 (Objective-bound redundancy without cardinality). *For nonnegative weights, adding (10)–(13) to CP0*

does not change its optimal value. This is an objective statement, not a claim that the metric inequalities leave the CP0 polyhedron unchanged.

Proof. Take an optimal CP0 solution and retain its label vector z . Set $x_{ij} = |z_i - z_j|$ for every pair. This extension satisfies the pairwise XOR hull and all metric inequalities because absolute distance on the line is a metric and, for $z_i \in [0, 1]$, $|z_i - z_j| + |z_i - z_k| + |z_j - z_k| \leq 2$. For fixed z , these are the smallest feasible pair values; nonnegative weights therefore ensure that the objective does not increase. Hence the metric-augmented optimum is no larger than the CP0 optimum. The reverse inequality follows because constraints were added. $\square$

Proposition 5 (Relaxation validity). *The B2S LP optimum is a valid lower bound on the $K = 2$ reference OPT.*

Proof. For binary labels, $x_{ij} = |z_i - z_j|$ satisfies (9). A balanced partition cuts $|F_0||F_1| = n^2/4$ complete-graph pairs, and each label triple satisfies (10)–(13). Hence every feasible integral bisection remains feasible in the relaxation, whose minimum cannot exceed OPT. $\square$

D. CertiCut Reference Implementation

H2 constructs feasible bisections from deterministic degree-based orderings and a spectral initialization based on the graph Laplacian’s Fiedler vector. Each candidate undergoes best-improving Kernighan–Lin-style swaps between fragments until no improving swap remains. The best candidate initializes UB; H2 does not establish a lower bound or certify optimality.

E. Backend-Independent Gap Semantics

Theorem 1 (Gap-to-overhead translation, exact arithmetic). *For any $K \geq 2$, feasible capacity bounds, and optimization backend, if $LB \leq OPT \leq UB$, where $UB = J(P_{\text{inc}})$ for a feasible incumbent of (5), then*

$$\frac{\Gamma(P_{\text{inc}})}{\Gamma^*} = \exp(UB - OPT) \leq \exp(UB - LB). \quad (14)$$

Proof. Equation (2) gives $\Gamma(P) = \exp(J(P))$. Since $OPT \geq LB$, exponentiation yields (14). $\square$

This result depends on the specified independent-QPD objective and valid global bounds, not on how a backend obtains them. Proposition 6 below establishes the required bound premise for the $K = 2$ CERTICUT reference search in exact arithmetic.

F. Reference Branch-and-Bound

The reference optimizer maintains open nodes in non-decreasing order of their LP bounds. It expands the best-bound node, updates the incumbent when an integral solution improves UB, and otherwise branches on the most fractional label. Child LPs reuse the root cut pool; any child whose lower bound is at least UB is pruned. After all LP solves associated with an expansion have completed, the algorithm reaches a *safe boundary* and computes

$$LB = \min\left(UB, \min_{N \in \mathcal{F}} LB_N\right), \quad (15)$$

Algorithm: Anytime CertiCut Reference Search

Input: QPD-weighted interaction graph G , capacity $q_{\max} = n/2$, soft deadline T .
Step 1. Compute initial feasible partition and upper bound: $(P_{\text{inc}}, UB) \leftarrow H2(G)$.
Step 2. Solve and strengthen root LP via B2S separation: $(LB_0, \mathcal{C}) \leftarrow B2S\text{-}ROOT(G)$. Freeze cut pool $\mathcal{C}$.
Step 3. Initialize search frontier $\mathcal{F}$ with root node.
Step 4. While $\mathcal{F} \neq \emptyset$ and elapsed time $< T$:

- Select node $N^* = \arg \min_{N \in \mathcal{F}} LB_N$.
- If N^* is integral and improves UB, update the incumbent.
- Otherwise, branch on the most fractional unfixed qubit; solve child LPs using $\mathcal{C}$; prune children with $LB_N \geq UB$.

Step 5. At each safe boundary, compute $LB = \min(UB, \min_{N \in \mathcal{F}} LB_N)$ and emit bound report.
Output: Best partition P_{inc} and bound report $(LB, UB, F = \exp(UB - LB))$.

Fig. 2. The CERTICUT reference anytime search for exact-balanced $K = 2$ partitioning.

where $\mathcal{F}$ denotes the open frontier. The additive log-domain gap and the corresponding multiplicative overhead factor are

$$\Delta = UB - LB, \quad F = \exp(\Delta). \quad (16)$$

Proposition 6 (Reference safe-boundary validity, exact arithmetic). *In exact arithmetic, every reference-search safe stopping boundary satisfies $LB \leq OPT \leq UB$.*

Proof. Feasibility of the incumbent gives $OPT \leq UB$. Every unexplored feasible solution belongs to a frontier subtree whose LP value lower-bounds all integral completions in that subtree. Equation (15) therefore gives $LB \leq OPT$. $\square$

Experimental deadlines are *soft*: the reference optimizer stops only at a safe boundary. We report the actual boundary time and never assign a later bound report to an earlier checkpoint. Propositions 5–6 are exact-arithmetic statements. Runtime results use floating-point HiGHS or SCIP bounds and are therefore labelled solver-tolerance quantities, not independently verified numerical proofs. A formal numerical bound would additionally require a verified conservative lower-bound procedure, such as dual-feasible recovery with outward rounding or rational/interval verification.

IV. EXPERIMENTAL METHODOLOGY

A. Experimental Protocol

All reported results come from an evidence corpus frozen on August 11–12, 2026. The environment uses Python 3.11.9, Qiskit 2.5.1, Qiskit Addon Cutting 0.10.0, Qiskit Aer 0.17.2, SciPy 1.17.1 [8], KaHIP 3.25 [9], MQT Bench 2.2.2 [10], PySCIPOpt 6.2.1, and SCIP 10.0 [11]. Experiments ran on Windows 11 Education (build 26200), an Intel Core i5-14400 (10 physical cores, 16 logical processors), and 31.69 GB RAM. Revision runners pin OpenMP, OpenBLAS, and MKL to one thread before numerical imports; the frozen legacy E1 data did not. We use an LP tolerance of 10^{-9} , fixed

random seeds, and fresh isolated processes for E1 deadlines. A manifest records exact versions, machine protocol, experiment counts, and SHA-256 hashes of raw results.

We evaluate ten experiment sets plus a separate representation-regret study; E8 is supplementary:

- **E1:** 420 CNOT-only balanced instances spanning seven graph families, six sizes $n \in \{16, 20, 24, 26, 32, 40\}$, and ten seeds per family–size cell. Each run uses a fresh process and a 60-s soft deadline.
- **E2:** 25 MQT Bench circuits [10] at $n \in \{8, 12, 16, 20, 24\}$, normalized to the $\{RZ, SX, X, CX\}$ basis.
- **E3:** 15 circuits from the same sources, preserving supported native gates such as RZZ and CZ .
- **E4:** 60 Qiskit automatic cut-finder runs under practical $Q_{\max}$ semantics. Because these runs use different fragment constraints and backjump budgets, we report them as contextual rather than equal-time baselines.
- **E5:** 120 exact balanced heterogeneous-QPD instances across five deterministic families, $n \in \{12, 14, 16\}$, and eight seeds per cell. Every circuit contains CX, CZ, iSWAP, $RZZ(\pi/8)$, $RZZ(\pi/4)$, and $RZZ(\pi/2)$; all costs are queried from the Qiskit QPD registry. Exhaustive enumeration computes both the canonical count-optimal and QPD-optimal plans.
- **E6:** 36 algorithm-derived MQT Bench circuits from six predeclared families and $n \in \{6, 8, 10, 12, 14, 16\}$. We selectively expand composites only until numeric one- and two-qubit semantic operations are exposed, without transpiling to a target or hardware basis. Each retained two-qubit occurrence is queried through Qiskit’s QPD registry; exhaustive enumeration evaluates both objectives.
- **E7:** 200 frozen heterogeneous-QPD records: five deterministic families, $n \in \{20, 24, 32, 40\}$, and ten seeds per cell. Exact-balanced $K = 2$ plans use the fixed cyclic gate assignment CX, CZ, iSWAP, $RZZ(\pi/8)$, $RZZ(\pi/4)$, and $RZZ(\pi/2)$. Independent 2-, 10-, and 60-s runs compare CERTICUT B2S-root+H2 with SCIP G0 (basic), G1 (cardinality), and G2 (full triangles); graph construction is excluded.
- **E8 (supplementary):** Three $n = 12$ E5 witnesses selected using modeled-overhead regret only, evaluated under fixed ideal-Aer shot budgets. This sanity study is summarized in the supplement because it does not establish a universal finite-shot predictor.
- **E9:** 144 capacitated K -way heterogeneous-QPD records: random-matching, community-matching, and weighted-repeat families; $n \in \{20, 32, 40, 60\}$; $K \in \{2, 3, 4, 5\}$; three fixed seeds per family–size– K cell; and exact near-balanced capacities. All use Qiskit’s independent-QPD registry, one numerical thread, and a 10-s SCIP limit. We report closure and solver-tolerance factors for every record; this is a replicated fixed benchmark matrix, not a workload-prevalence estimate.

TABLE I
EXPERIMENT-TO-QUESTION MAP. E7 AND E9 ARE THE PRIMARY BACKEND AND K -WAY SCALABILITY STUDIES; E8 IS SUPPLEMENTARY.

Set	Question
E1	Legacy CNOT-only characterization of the reference implementation.
E2/E3	Dependence on specified CX-normalized versus native representation.
E4	Narrow overlap with Qiskit’s practical multi-fragment cutter.
E5/E6	Tie-safe count-versus-QPD reversals, constructed and algorithm-derived.
E7	Which backend gives the strongest $K = 2$ anytime bounds on heterogeneous-QPD instances?
E8	Does lower modeled overhead correspond to lower finite-shot error at equal budget?
E9	How do replicated $K \in \{2, 3, 4, 5\}$ closure and bound factors vary through $n = 60$?
E10	Does the K -way boundary persist on algorithm-derived native-QPD circuits?
Rep.-regret	Does a semantically equivalent circuit representation alter the independent-QPD-optimal placement, and when is that placement guaranteed stable?

- **E10:** 36 algorithm-derived native-QPD records from MQT Bench QAOA, QFT, exact QPE, and Draper-QFT-adder families at $n \in \{20, 24, 32\}$ and $K \in \{2, 3, 4\}$, plus three explicitly recorded VQE ingestion failures due to unsupported native two-qubit QPD gates. The same 10-s SCIP protocol applies.
- **Representation-regret study:** 95 paired representation records across eight MQT Bench families (QAOA, QFT, QPE, BV, Grover, VQE, GHZ, DJ) at $n \in \{4, 6, 8, 10, 12, 14, 16\}$ and $K \in \{2, 3\}$. Both representations originate from the same deterministic logical source circuit over identical logical qubits with no routing map. For $n \leq 10$, unitary equivalence up to global phase is verified explicitly ($< 10^{-13}$ max entrywise deviation); for $n > 10$, identical source SHA-256 fingerprints are verified. Small instances ($n \leq 10$) use exhaustive enumeration over the complete feasible partition space $\mathcal{F}$. Medium instances ($n \in \{12, 14, 16\}$) use a two-stage lexicographic MIP ($\min J_b(P)$ s.t. $J_a(P) \leq J_a^* + \tau_{\text{opt}}$) evaluated across $\tau_{\text{opt}} \in \{10^{-8}, 10^{-9}, 10^{-10}\}$.

Table I maps each experiment to its research question and prevents representation, exact-oracle, and scalability evidence from being conflated.

B. Metrics and Baselines

We report solver-tolerance gap closure, bound-report availability, factor F , optimizer and end-to-end runtime, and peak resident memory. Every E1 checkpoint uses the full denominator of 420; a run that has not reached a safe boundary is counted as unavailable. E7 reports F , threshold counts, closure, and incumbent agreement. E9/E10 report status, incumbent, primal/dual bounds, and F for every record. These corpora are fixed designs rather than IID samples, so proportions are descriptive rather than population-prevalence inference.

KaHIP supplies fast feasible bisections for CNOT-only instances but no optimality certificate. For E9, a capacity-exact multistart weighted greedy-swap heuristic is the direct K-way baseline. It uses the same aggregate log-QPD weights and exact capacities, returns a feasible plan P_H , and has no bound of its own. CertiCut’s SCIP lower bound evaluates it as $\Gamma(P_H)/\Gamma^* \leq \exp(J(P_H) - \text{LB})$. Qiskit’s automatic cut finder [5] uses a different width-constrained gate/wire-cut model and remains contextual.

C. E7 Backend Protocol

E7 is the primary backend comparison. All methods run single-threaded, pinning `OMP_NUM_THREADS`, `OPENBLAS_NUM_THREADS`, and `MKL_NUM_THREADS` to one before numerical-library import. CERTICUT uses its 10^{-9} solver-tolerance protocol; SCIP uses its documented numerical feasibility tolerance. At each independent wall-clock checkpoint, all methods emit common UB, LB, and $F = \exp(\text{UB} - \text{LB})$ fields. These are solver-tolerance numerical quantities, not independently verified numerical proofs. The frozen protocol, raw records, checkpoint requirements, and SHA-256 hashes are included in the artifact.

V. EXPERIMENTAL RESULTS

A. Gate Costs Change the Optimization Target

Unweighted cut count can rank feasible plans incorrectly. Two iSWAP cuts have modeled overhead $49^2 = 2,401$, whereas three CNOT cuts cost $9^3 = 729$; the three-cut plan is cheaper. Likewise, two CS cuts cost $(3 + 2\sqrt{2})^2 \approx 33.97$, compared with 81 for two CNOT cuts. Equation (3) preserves these gate-dependent distinctions.

E5 tests this distinction under controlled heterogeneity. It uses validated Qiskit costs for CX, CZ, iSWAP, and three nonzero RZZ angles on 120 deterministic circuits; no operation has unit QPD overhead. To avoid tie-breaking artifacts, we compare $\min_{P: c(P)=c^*} J(P)$, the best QPD objective among all minimum-cut partitions, with $\min_P J(P)$; both values are obtained by exhaustive enumeration. This stronger regret is strict in 16/120 cases (13.3%). Its conditional factor has median 5.30, 90th percentile 27.47, and maximum 181.11. Every strict reversal requires more cuts under a QPD-optimal plan, with median count increase one and maximum three. Across the 16 reversals, QPD-optimal plans cut 28 fewer iSWAP gates while accepting 47 additional lower-overhead gates (CX, CZ, and RZZ). All 120 QPD optimization runs close numerically within the 30-s deadline; 90 close at the B2S root.

E6 tests whether this mechanism occurs without constructed gate palettes. Its predeclared protocol fixes six algorithm-derived MQT Bench families [10], six even sizes, the eligibility criterion, and the decision rule before inspecting outcomes. It selectively expands only operations with arity above two, preserves numeric one- and two-qubit semantic operations, never transpiles to a target basis, and queries every retained two-qubit occurrence via

TABLE II
IDENTITY-FREE HETEROGENEOUS-QPD COMPANION CORPUS (E5). ALL COSTS COME FROM QISKIT’S INDEPENDENT-QPD REGISTRY.

Metric	Value
Instances	120 (5 families; $n \in \{12, 14, 16\}$)
Strict reversals	16 (13.3%)
Median / p90 factor	$5.30 \times / 27.47 \times$
Extra cuts, median / max	1 / 3
Avoided iSWAP cuts	28
Numerically closed	120/120

TABLE III
E6 ALGORITHM-DERIVED MQT BENCH AUDIT. “MAX FACTOR” IS DESCRIPTIVE OVER SIX FIXED SIZES PER FAMILY, NOT A POPULATION ESTIMATE.

Family	Eligible	Reversals	Max factor
Draper QFT adder	6/6	4/6	1.04×10^6
HHL	6/6	0/6	1
Exact QPE	6/6	2/6	4.03
Inexact QPE	6/6	0/6	1
QFT	6/6	0/6	1
Entangled QFT	6/6	0/6	1

`QPDbasis.from_instruction`. All 36 circuits are eligible: none has an unsupported or unit-overhead two-qubit operation, and each has at least two positive QPD-cost classes. Exhaustive tie-safe enumeration finds six strict reversals: four among six Draper QFT adders and two among six exact-QPE circuits. HHL, inexact QPE, QFT, and entangled-QFT supply 24 negative results despite heterogeneous costs. Thus heterogeneity is necessary for the observed mechanism but is not sufficient to change the optimum.

For E6, the count objective $c(P)$ and QPD objective $J(P)$ define the tie-safe regret

$$\Delta_{E6} = \min_{P: c(P)=c^*} J(P) - \min_P J(P), \quad c^* = \min_P c(P), \quad (17)$$

and a strict reversal requires $\Delta_{E6} > 10^{-10}$. Fixing qubit zero removes complementary-label symmetry, leaving $\binom{n-1}{n/2-1}$ balanced partitions; E6 exhaustively evaluates all of them, from 10 at $n = 6$ through 6,435 at $n = 16$.

The largest witness is the Draper QFT adder at $n = 16$: the best count-optimal plan has 36 cuts and $J = 37.968133$, whereas the QPD-optimal plan has 42 cuts and $J = 24.111328$, yielding a factor 1.042×10^6 . The QPD plan avoids expensive $CP(\pi/2)$ and $CP(\pi)$ crossings by accepting more small-angle controlled phases. Direct occurrence-level sums and independently aggregated pair weights agree within 10^{-10} for both plans in every reversal record. This is existence evidence on algorithm-derived benchmark circuits, not a prevalence estimate over workloads or a hardware-routing result.

B. Mature MIP Is the Strongest Evaluated Backend

E7 makes the backend result explicit. Table V reports all 200 frozen heterogeneous-QPD instances at each independent checkpoint. **SCIP G0 is the strongest practical backend**

TABLE IV

GROUPED NONZERO-DELTA GATE-CLASS ACCOUNTING FOR THE DRAPER $n = 16$ WITNESS. THE GROUPS SUM TO +6 CUTS AND $J_{\text{QPD}}^* - J_{\text{count}}^* = -13.856805$; FULL PER-ANGLE ACCOUNTING IS IN THE ANONYMIZED ARTIFACT.

CP angle class	Count	QPD	Δc
$-\pi/64$ through $-\pi/4$	5	13	+8
$\pi/64$ through $\pi/16$	6	15	+9
$\pi/4$	5	2	-3
$\pi/2$	6	2	-4
π	6	2	-4
Total	28	34	+6

TABLE V

E7 FROZEN HETEROGENEOUS-QPD BACKEND COMPARISON. ENTRIES ARE COUNTS WITH $F \leq 1.01$; THEY EQUAL SOLVER-TOLERANCE CLOSURES HERE. ALL FACTORS AND CLOSURES ARE SOLVER-TOLERANCE NUMERICAL QUANTITIES.

Method	2 s	10 s	60 s	60 s UB match G0
CERTICUT	160/200	184/200	199/200	199/200
SCIP G0	199/200	200/200	200/200	reference
SCIP G1	111/200	186/200	196/200	198/200
SCIP G2	108/200	158/200	183/200	189/200

among the evaluated methods. It reaches solver-tolerance closure on 199/200 instances at 2 s and all 200 by 10 s. The specialized CERTICUT reference implementation closes 160/200, 184/200, and 199/200 at 2, 10, and 60 s, respectively. The $F \leq 1.01$ and solver-tolerance-closure counts happen to coincide on this corpus. These results do not support a specialized-solver performance advantage. Instead, they show that the log-domain model-factor semantics can be instantiated by either the reference implementation or a mature MIP solver.

At 60 s, CERTICUT matches the G0 incumbent on 199/200 records; its open case is a dense-shuffled $n = 40$ instance with $F = 6.1262$. Its incumbent exceeds G0's by 1.00041 log-objective units, equivalent to a modeled-overhead ratio of $\exp(1.00041) \approx 2.72$. Statically adding cardinality or full triangles to SCIP does not improve its end-to-end performance on this corpus. This does not contradict the specialized root study below: it concerns LP objective strength under the reference formulation, whereas E7 measures mature-solver branch-and-bound behavior under fixed time budgets.

C. Capacitated K-Way Heterogeneous Scaling

E9 evaluates the core formulation beyond bisection. Table VI and Fig. 3 summarize all 144 heterogeneous-QPD runs, including open cases. SCIP reaches solver-tolerance closure on 101/144 runs within 10 s. The boundary is strongly graph and seed dependent. Community matching closes all three seeds through $(K, n) = (3, 60)$ and $(4, 40)$; weighted repeat closes all three through $(3, 60)$ and $(4, 40)$. Random matching is the hard family: all three $K = 2, n = 60$ runs close, whereas every $K \geq 3, n \geq 40$ run remains open.

Each E9 circuit contains four interaction rounds. Random matching draws a fresh perfect matching per round. Commu-

TABLE VI

E9 CAPACITATED K -WAY HETEROGENEOUS-QPD SCALING, THREE FIXED SEEDS PER CELL AND 10-S SCIP LIMIT. "CLOSED" MEANS SOLVER-TOLERANCE CLOSURE. OPEN RECORDS RETAIN FINITE FACTORS; THIS FIXED MATRIX IS NOT A WORKLOAD-PREVALENCE ESTIMATE.

Family	Closed	All-seed closed	Boundary
Random matching	24/48	$K2, n60; K4, n20$	$K \geq 3, n \geq 40$
Community matching	37/48	$K3, n60; K4, n40$	$K5, n \geq 32$
Weighted repeat	40/48	$K3, n60; K4, n40$	$K \geq 4, n = 60$
Total	101/144	$n = 60, K = 3$	family/seed dependent

nity matching draws separate within-half matchings and inserts two fixed bridge pairs per round. Weighted repeat uses a ring backbone and diametric chords, repeating each pair one to four times. Gate occurrences cycle deterministically through CX, CZ, iSWAP, $RZZ(\pi/8)$, $RZZ(\pi/4)$, and $RZZ(\pi/2)$ in circuit order; this assignment is seed-independent. The seed controls only matching order or repeat multiplicities.

The factors quantify why closure alone is insufficient. Across the three random-matching $n = 32, K = 5$ runs, the median open factor is $10^{15.47} \approx 2.9 \times 10^{15}$; at $n = 60, K = 4$ it is $10^{44.84} \approx 6.9 \times 10^{44}$. These values state that the corresponding incumbents have no useful solver-tolerance multiplicative model-overhead guarantee at this deadline. Conversely, their explicit reporting prevents an unproven incumbent from being presented as an optimum. Thus E9 establishes a practical baseline and a transparent computational boundary, not universal large- K scalability.

The exact-capacity comparison also expresses the fragment-width/overhead tradeoff directly: increasing K reduces the maximum fragment size while generally forcing more independent-QPD boundaries. Fig. 4 plots this relation for closed runs using the median closed $\log \Gamma^*$ in each cell. The homogeneous small matrix is only an implementation check; E9 supplies replicated heterogeneous sampling-aware scaling evidence. The three seeds improve stability assessment but remain a fixed benchmark design rather than a workload distribution. The deterministic cost cycle is seed-independent, but residual ordering effects are a limitation rather than a proof of zero topology-cost correlation.

D. Direct Capacity-Exact K-Way Heuristic Baseline

The weighted K-way greedy-swap baseline uses the same aggregate log-QPD weights and exact capacity vector as E9, then performs best-improving inter-fragment swaps. It returns every E9 feasible partition in a median 21.6 ms, but has no internal lower bound. On the 101 SCIP-closed records, CertiCut's closed lower bounds evaluate its exact multiplicative regret: 23 heuristic partitions are optimal, the median factor is 28.05, the 90th percentile is 3.32×10^{10} , and the maximum is 5.80×10^{20} . For the remaining 43 time-limited records, the same anytime expression remains valid but loose: its median upper factor is 5.79×10^{15} , its 90th percentile is 5.81×10^{30} , and its maximum is 9.97×10^{31} . This comparison is intentionally not a reproduction of Nakamura et al. or Fröhler et al.; it is a capacity-exact weighted K-way heuristic under precisely

TABLE VII

E9 CAPACITY-EXACT WEIGHTED K-WAY GREEDY-SWAP BASELINE, EVALUATED WITH CERTICUT SCIP BOUNDS. FACTORS USE ONLY THE 101 SCIP-CLOSED RECORDS AND ARE EXACT UP TO SOLVER TOLERANCE.

Metric	Value
Feasible heuristic records	144/144
Median heuristic runtime	21.6 ms
SCIP-closed comparison records	101
Heuristic-optimal records	23/101
Median / p90 regret factor	28.05 / 3.32×10^{10}
Maximum regret factor	5.80×10^{20}

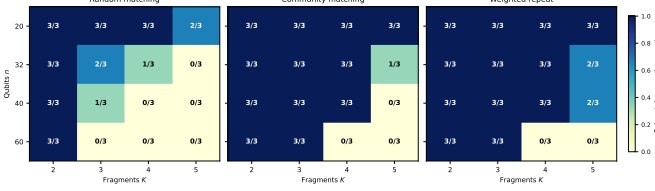


Fig. 3. E9 solver-tolerance closure counts under a 10-s SCIP limit. Each heatmap cell reports closed runs out of three fixed seeds. The boundary depends materially on graph family, K , and seed.

the same objective. It demonstrates the intended use of (14): a scalable heuristic incumbent can be assigned a resource-interpretable regret factor whenever SCIP supplies a lower bound.

E. Algorithm-Derived K-Way Boundary

E10 applies the same native independent-QPD model to algorithm-derived MQT Bench circuits. Among 36 valid QAOA, QFT, exact-QPE, and Draper-QFT-adder records, SCIP closes 15 within 10 s: 11/12 at $K = 2$, 3/12 at $K = 3$, and 1/12 at $K = 4$. The six Draper-adder records provide the largest structured positive subset, closing all $n = 20$ plans and both $K = 2, 3$ plans at $n = 24$. Three VQE candidates are retained as explicit ingestion failures because their native two-qubit operations lack a Qiskit QPD basis. Thus the E9 boundary is not solely a synthetic-graph artifact, while E10’s small fixed matrix does not estimate algorithm-family prevalence.

F. Finite-Shot Scope

Supplementary E8 fixes an equal total ideal-Aer shot budget on three predeclared $n = 12$ witnesses. The strong $719.66 \times$ modeled reversal reduces RMSE for both observables, whereas a moderate $3.27 \times$ reversal is mixed and a no-reversal control is identical under shared seeds. Hence independent-QPD overhead can be operationally informative but is not claimed as a universal finite-shot RMSE predictor. Full witness identities, signed-estimator behavior, and bootstrap intervals remain in the artifact supplement.

G. Artifact Availability

The accompanying review package contains the reference implementation, frozen raw results, experiment scripts, and

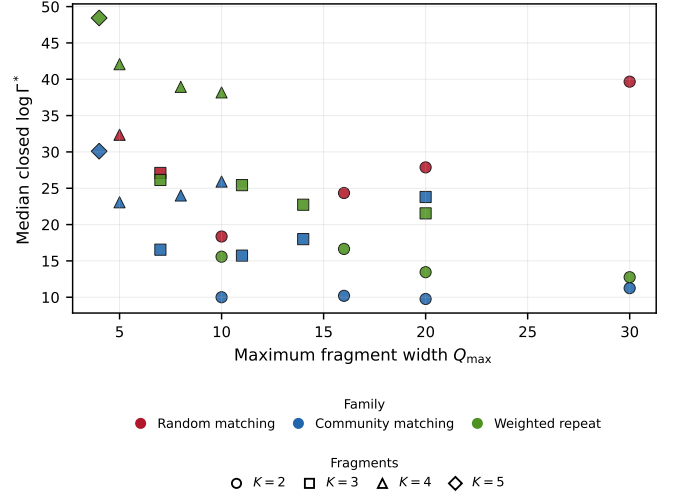


Fig. 4. E9 width-overhead tradeoff restricted to cells closed for all three seeds. Colors denote families; markers denote K . Each point is the median $\log \Gamma^*$ for its cell. This avoids conditioning the plot on solver-successful seeds.

SHA-256 manifests. It includes complete E1/E5/E7 generators, deterministic H2 ordering and swap rules, B2S cut policy/tolerance, branching rules, solver options, E6 gate-occurrence accounting, and frozen E8 plan and seed records. For the core K -way evidence it additionally includes the E9 generator, exact near-balanced capacity vectors, fixed gate-cost assignment order, all 144 raw records, the capacity-exact greedy-swap baseline with its restart/seed policy and 144 regret records, the E10 MQT ingestion runner, all 36 valid E10 records, and the three recorded unsupported-VQE ingestion failures.

H. Cardinality and Metrics Are Complementary

To isolate the source of root-bound strength, we compare four relaxations with identical complete-pair variables and the same LP backend: CP0; CP0 plus cardinality (C); CP0 plus separated metric inequalities (T); and their combination (CT). The study contains 24 circuit-derived instances from the random, dense, weighted-random, and noisy-community families at $n \in \{16, 24\}$, with three seeds per cell. C and T each reproduce the CP0 bound on every instance. Proposition 4 explains the CP0+T equality structurally. CT improves all 24 bounds and closes 12 roots, showing empirically that cardinality activates the metric system strongly on this corpus by supplying global cut mass that the metric geometry must distribute.

Separating all violated triangles at $n \in \{32, 40\}$ exceeded the 10-min audit budget before a complete matrix was obtained; we therefore report no tail result for this study.

I. Strengthening Improves Reference-Search Closure

Table IX reports a controlled 90-instance ablation with a common 100-node budget. A0 closes 41 gaps, regardless of whether H2 is enabled. Both B2S-root configurations close all 90 with a median of one explored node. Within this

TABLE VIII

ROOT-RELAXATION DECOMPOSITION ON 24 CIRCUIT-DERIVED INSTANCES. C AND T SEPARATELY MATCH CP0 ON EVERY INSTANCE; CT DENOTES THEIR COMBINATION. GAPS ARE IN LOG-OVERHEAD UNITS.

Root formulation	Exact	Med. gap	p90 gap	Med. time
CP0	0/24	26.2935	52.4468	0.3211 s
CP0 + C	0/24	26.2935	52.4468	0.3429 s
CP0 + T	0/24	26.2935	52.4468	0.2832 s
CP0 + C + T	12/24	0.2848	3.8761	2.3222 s

TABLE IX

CONTROLLED ABLATION ON 90 SYNTHETIC BALANCED INSTANCES UNDER A 100-NODE BUDGET. “CLOSED” MEANS SOLVER-TOLERANCE GAP AT MOST 10^{-9} ; NODE-LIMITED RUNS MAY ALSO BE CLOSED.

Configuration	Closed	Node-lim.	Med. nodes	Root / Tree (s)
A0 + H0 (basic)	41/90	56	100	0.000 / 2.059
A0 + H2	41/90	52	100	0.000 / 2.072
B2S-root + H0	90/90	0	1	0.260 / 0.060
B2S-root + H2	90/90	0	1	0.256 / 0.063

protocol, the complete-pair strengthening, rather than incumbent construction, explains the change. Because A0 and B2S use different pair spaces, this ablation measures the package rather than only C and T; Table VIII provides the matched-representation isolation. The SCIP audit above further shows that neither result establishes general solver superiority.

J. Legacy Reference Scope

The artifact supplement retains E1, the 420-instance CNOT-only characterization of the $K = 2$ B2S reference implementation. E7 is the matched production-backend result: basic SCIP closes 200/200 heterogeneous $K = 2$ records within 10 s and outperforms the specialized reference. We therefore use the legacy material only to document B2S polyhedral behavior, not as a central K-way scalability claim.

K. Representation Sensitivity and Optimal-Placement Regret

A circuit representation change affects not only absolute independent-QPD resource costs but can alter the optimal partition placement itself. We quantify this decision effect using tie-safe cross-representation placement regret and formalize conditions for placement stability.

1) *Tie-Safe Placement Regret*: Consider a logical circuit on a fixed set of logical qubits $V = \{0, \dots, n-1\}$ evaluated over a common feasible partition space $\mathcal{F}$ (defined by declared fragment count K and capacity bounds). Let a and b be two semantically equivalent representations with interaction edge sets E_a and E_b . We define the common edge universe $E = E_a \cup E_b$ and zero-pad the log-weights: $w_e^{(a)} = 0$ for $e \notin E_a$ and $w_e^{(b)} = 0$ for $e \notin E_b$.

For any partition $P \in \mathcal{F}$, the log objectives are $J_a(P) = (w^{(a)})^\top z(P)$ and $J_b(P) = (w^{(b)})^\top z(P)$, where $z(P) \in \{0, 1\}^{|E|}$ is the cut indicator vector. Let $\mathcal{A}_a = \arg \min_{P \in \mathcal{F}} J_a(P)$ and $J_b^* = \min_{P \in \mathcal{F}} J_b(P)$. The *tie-safe*

cross-representation regret and corresponding sampling overhead factor are:

$$\Delta_{a \rightarrow b} = \min_{P \in \mathcal{A}_a} J_b(P) - J_b^*, \quad R_{a \rightarrow b} = \exp(\Delta_{a \rightarrow b}). \quad (18)$$

The minimum makes the metric tie-safe and conservative: it reports positive regret only when every a -optimal placement is suboptimal under b ; it is not a worst-case arbitrary-tie-breaking metric. When $R_{a \rightarrow b} = 1$, at least one optimal partition under representation a remains optimal under b ($\mathcal{A}_a \cap \mathcal{A}_b \neq \emptyset$). When $R_{a \rightarrow b} > 1$, every optimal partition under a is strictly suboptimal under b . In general, $R_{a \rightarrow b} \neq R_{b \rightarrow a}$.

2) *Theoretical Bound and Stability Theorem*: The following perturbation statements are exact model-level results; the medium-tier numerical evaluations below are solver-tolerance quantities. Let $\delta = w^{(b)} - w^{(a)} \in \mathbb{R}^{|E|}$ be the interaction-weight perturbation vector over the edge universe E .

Proposition 7 (Representation-Regret Bound). *For any common feasible partition space $\mathcal{F}$, $0 \leq \Delta_{a \rightarrow b} \leq \|\delta\|_1$, and $R_{a \rightarrow b} \leq \exp(\|\delta\|_1)$.*

Proof. Let $P_a \in \arg \min_{P \in \mathcal{A}_a} J_b(P)$ and $P_b \in \mathcal{A}_b$. Then $J_b(P_a) - J_b(P_b) = [J_a(P_a) - J_a(P_b)] + \delta^\top [z(P_a) - z(P_b)]$. Because P_a is J_a -optimal, $J_a(P_a) - J_a(P_b) \leq 0$. Since $[z(P_a) - z(P_b)]_e \in \{-1, 0, 1\}$, we obtain $\Delta_{a \rightarrow b} \leq \delta^\top [z(P_a) - z(P_b)] \leq \|\delta\|_1$. $\square$

Let $m_a = \min_{P \in \mathcal{F} \setminus \mathcal{A}_a} [J_a(P) - J_a^*]$ denote the optimality margin of representation a , with the convention $m_a = +\infty$ if $\mathcal{F} \setminus \mathcal{A}_a = \emptyset$.

Theorem 2 (Representation Stability). *If $m_a > \|\delta\|_1$, then $\mathcal{A}_b \subseteq \mathcal{A}_a$, and consequently $R_{a \rightarrow b} = 1$.*

Proof. For any $P \in \mathcal{A}_a$ and $Q \in \mathcal{F} \setminus \mathcal{A}_a$, $J_a(Q) - J_a(P) \geq m_a$. The perturbation changes the objective difference by at most $|\delta^\top [z(Q) - z(P)]| \leq \|\delta\|_1$. If $m_a > \|\delta\|_1$, then $J_b(Q) - J_b(P) \geq m_a - \|\delta\|_1 > 0$, so no non-optimal partition $Q \notin \mathcal{A}_a$ can achieve a smaller or equal J_b objective than P . Thus $\mathcal{A}_b \subseteq \mathcal{A}_a$, guaranteeing $R_{a \rightarrow b} = 1$. $\square$

Defining $\kappa_{a \rightarrow b} = \|\delta\|_1 / m_a$, $\kappa_{a \rightarrow b} < 1$ provides a sufficient condition for placement stability.

Corollary 1 (Positive-Scaling Invariance). *If $w^{(b)} = c w^{(a)}$ for $c > 0$, then $\mathcal{A}_a = \mathcal{A}_b$ and $R_{a \rightarrow b} = R_{b \rightarrow a} = 1$.*

Proof. For any $P \in \mathcal{F}$, $J_b(P) = c J_a(P)$. Since $c > 0$, the scalar multiplier preserves exact partition objective ordering, so $\arg \min J_a = \arg \min J_b$ and $R_{a \rightarrow b} = R_{b \rightarrow a} = 1$. $\square$

3) *Tie-Safe Evaluation Protocol and Empirical Results*: To evaluate placement regret without arbitrary solver tie-breaking artifacts, we employ a two-stage lexicographic MIP for larger instances ($n \in \{12, 14, 16\}$): Stage 1 computes J_a^* and J_b^* ; Stage 2 minimizes $J_b(P)$ subject to $J_a(P) \leq J_a^* + \tau_{\text{opt}}$. We repeat Stage 2 with a and b exchanged to obtain the reverse-direction regret $\Delta_{b \rightarrow a}$. We evaluate across $\tau_{\text{opt}} \in \{10^{-8}, 10^{-9}, 10^{-10}\}$ using PySCIPopt 6.2.1 and Qiskit 2.5.1 / Addon Cutting 0.10.0. Exact enumeration is used for $n \leq 10$.

Table X summarizes the findings:

TABLE X

TIE-SAFE CROSS-REPRESENTATION PLACEMENT REGRET ACROSS 95 PAIRED REPRESENTATION RECORDS (Δ IN NATURAL-LOG OVERHEAD UNITS). RECORDS WITH $n \leq 10$ ARE EXHAUSTIVE; $n \geq 12$ RECORDS USE SOLVER-TOLERANCE LEXICOGRAPHIC MIP. ALL REPORTED ROW-LEVEL OUTCOMES ARE UNCHANGED ACROSS TESTED τ_{opt} VALUES.

Family	Records	Tie-Safe Rev.	Max $\Delta_{\text{CX} \rightarrow \text{native}}$	Max $\Delta_{\text{native} \rightarrow \text{CX}}$
QAOA	12	0	0.0	0.0
QFT	12	0	0.0	0.0
QPE-exact	12	3 (25%)	3.31	6.59 ($\log_{10} R = 2.86$)
Grover	11	5 (45%)	70.31 ($\log_{10} R = 30.5$)	144.57 ($\log_{10} R = 62.8$)
BV	12	0	0.0	0.0
Controls (VQE,GHZ,DJ)	36	0	0.0	0.0

- 1) **Confirmed Placement Reversals:** 8/95 (8.4%) records exhibit strict tie-safe placement reversal on QPE and Grover circuits. For QPE at $n = 14, K = 2$, $\Delta_{\text{native} \rightarrow \text{CX}} = 6.5917$ ($\log_{10} R = 2.86$, $R = 729.0$), indicating a 3-order-of-magnitude placement regret. Grover at $n = 16, K = 3$ reaches $\Delta = 144.57$ ($\log_{10} R = 62.8$). All reversals are consistent across all tested τ_{opt} thresholds. All medium-tier Stage-1 and Stage-2 problems reported as reversals reached solver-tolerance closure. One predeclared Grover record ($n = 16, K = 2$) was excluded due to transpiler memory limit during composite decomposition.
- 2) **Elimination of Single-Solve Artifacts:** Single-solve SCIP evaluations initially reported 6 QFT reversals with R up to 27,741. Under the two-stage tie-safe MIP, all 6 QFT cases resolve to $R = 1.0$ ($\Delta = 0.0$). The QFT cases admit alternative CX-optimal placements that are also native-optimal, showing that arbitrary single-optimum cross-evaluation can create false-positive placement regret.
- 3) **Collinearity and Theorem Verification:** QAOA exhibits $R = 1.0$ across all sizes despite a $2\times$ cost shift ($J_{\text{CX}}^* = 158.2$ vs $J_{\text{native}}^* = 79.1$ at $n = 16$), as predicted by Positive-Scaling Invariance. Among exact-tier records satisfying $\kappa < 1$ (including zero-perturbation controls), all 25 satisfy $R = 1.0$ as required by Theorem 2.

L. Comparison with Practical Automatic Cutting

E4 contains 60 Qiskit cut-finder records for QAOA, BV, and VQE at $n \in \{8, 12, 16\}$. Qiskit marks 24 records as “minimum reached.” For these records, the best balanced- $K = 2$ objective equals the reported practical objective within numerical tolerance: the median, 90th percentile, and maximum restriction factors are 1.0, 1.0, and 1.0000000000000007. The agreement includes BV cases where Qiskit returns more than two fragments.

This is a narrow overlap result, not evidence of equivalence between balanced bisection and unconstrained practical cutting. It covers three circuit families, three sizes, one QPD model, and only records carrying Qiskit’s own optimality flag.

M. Exact Reconstruction Sanity Checks

We pass two selected partitions through Qiskit’s complete gate-cutting pipeline: QPD replacement, fragment-experiment

TABLE XI

EXACT RECONSTRUCTION CHECK. PREDICTED AND QISKIT-DERIVED OVERHEADS AGREE; “ABS. ERROR” COMPARES RECONSTRUCTED AND UNCUT OBSERVABLES.

Circuit	Cuts	Pred. Γ	Exec. Γ	Abs. error
QAOA $n=6$, native RZZ	2	33.9706	33.9706	1.39×10^{-17}
VQE $n=6$, CX	1	9	9	5.55×10^{-17}

generation, exact statevector evaluation of every subexperiment, and observable reconstruction. For a six-qubit QAOA circuit, two native RZZ cuts give predicted and executed overhead $\Gamma = 33.9706$ and absolute reconstruction error 1.39×10^{-17} . For a six-qubit VQE circuit, one CX cut gives overhead 9 and error 5.55×10^{-17} . In both cases, the reconstructed expectation agrees with the uncut statevector to machine precision.

This small-scale test checks cost accounting and pipeline compatibility. It does not evaluate finite-shot variance: every QPD basis term is evaluated exactly. Supplementary E8 supplies the separate finite-shot sanity study.

VI. RELATED WORK

Peng et al. [2] introduced circuit cutting for evaluating circuits wider than the available quantum device, building on quasiprobability simulation [3]. Mitarai and Fujii developed virtual-gate constructions and overhead bounds for nonlocal channel simulation [6], [12]. Harrow and Lowe characterize cut costs through product extent and give cost bounds for spacelike and timelike cuts [13]. CutQC couples automatic partitioning with classical recombination [4], while Qiskit Addon Cutting implements gate-level QPD experiment generation and reconstruction [5]. Complementary directions include maximum-likelihood fragment tomography [14], fragmentation for error reduction [15], operator backpropagation [16], and sampling bounds beyond bipartitions [17].

Nakamura, Satoh, and Sanji [17] study sampling bounds and scalable partitioning beyond bipartitions. Their objective and cut-placement setting address multi-fragment cutting directly. CERTICUT instead fixes an independent gate-QPD registry and solves a capacitated graph-partitioning MIP, translating a valid global log-objective gap into a multiplicative model-overhead factor. Fröhler et al. [18] propose a broader scalable heuristic framework for combined gate cuts, wire cuts, and joint gate groups. Their cut-mechanism scope is broader; CERTICUT is narrower and focuses on the fixed independent-QPD objective and solver-tolerance global-bound semantics.

Brandhofer, Polian, and Krsulich [19] optimize richer circuit-knitting partitions that jointly select gate and wire cuts, ancilla insertions, and classical communication under gate-dependent execution and sampling costs. Idan et al. [20] establish NP-completeness for general cut placement and use SMT for bounded-size fragments. Tamura et al. [21] investigate partition-dependent multi-qubit gate decomposition strategies, including ancilla-assisted cut optimizations. These works are broader along the axis of cut mechanics and exe-

cution resources. CERTICUT generalizes its earlier bisection formulation along different axes: declared fragment count, capacity constraints, and translation of a global independent-QPD optimization gap into a multiplicative model-overhead factor. Furthermore, while Tamura et al. optimize gate decomposition strategies, CERTICUT holds the logical wire set and feasible partition space fixed and analyzes how semantically equivalent representations perturb interaction weights to induce or preserve optimal placement decisions. In distributed quantum computing, multilevel temporal-hypergraph partitioning instead optimizes teleportation and entanglement cost [22]; it is a distinct resource model from circuit cutting.

CERTICUT deliberately fixes a per-gate independent-QPD registry, then optimizes capacitated K -way placement of those costs across a partition. Weighted graph partitioning, branch-and-bound bounds, fixed-capacity cardinality, and metric inequalities are established optimization machinery [7], [23]; we do not claim them as new. Our contribution is the model specialization and its resource-interpretable bound semantics, controlled and algorithm-derived objective-reversal evidence, and transparent backend evaluation. E7 does not claim a specialized branch-and-bound procedure dominates general-purpose optimization; it shows the opposite on the evaluated $K = 2$ corpus. Rather, the exact model reduction and factor semantics are directly usable by a mature optimizer. Multilevel partitioners such as KaHIP [9] provide strong feasible bisections but do not supply the same global bound.

Supplementary E8 does not establish globally optimal decomposition cost. It tests only whether plan differences under the specified independent-QPD objective can have finite-shot consequences in the evaluated ideal-simulator implementation.

VII. LIMITATIONS AND CONCLUSION

A. Limitations

The core formulation is restricted to gate-cut partitions under a fixed circuit representation, logical interaction graph, and independent per-gate QPD registry. Each optimization instance fixes one circuit representation; the representation-regret study compares separate fixed-representation instances but does not jointly optimize over representation choice and partition placement. It supports arbitrary declared K and capacity bounds but does not optimize wire cuts, exact global routing, hardware noise, or an end-to-end physical execution objective. Joint QPD can reduce sampling overhead when compatible cut gates admit a collective decomposition, but its structural cost semantics lie outside the independent per-gate model optimized here; general joint-aware partition optimization remains future work [24]. The sparse K -way LP relaxation is weak; the tested dense cardinality/metric extension improves neither closure nor runtime at the reported scale. E9 uses three deterministic seeds per family-size- K cell; this replication exposes seed sensitivity but remains a fixed matrix rather than a workload-prevalence estimate. Random-matching $K \geq 3$ instances retain weak factors at 10 s. E10 broadens the evidence to a small algorithm-derived matrix but remains non-prevalence evidence. Supplementary E8 evaluates finite-shot

behavior rather than modeling it. Finally, empirical optimization bounds are obtained from floating-point solvers. They are solver-tolerance quantities and have not been independently verified using directed rounding, rational reconstruction, or another rigorous lower-bound procedure.

The frozen manifest records software versions, solver tolerance, seeds, experiment counts, and SHA-256 hashes. Reproducing the reported numbers requires this protocol; changing the solver stack requires regenerating the evidence.

B. Conclusion

This work organizes capacitated circuit-cutting optimization into two core contributions. First, for capacitated K -way gate partitioning under an independent per-gate QPD model, we show that the log-overhead objective reduces exactly to weighted graph partitioning. This reduction yields backend-independent model-overhead bound semantics: in exact arithmetic, any global primal/lower-bound pair (UB, LB) guarantees an incumbent sampling overhead bound $\Gamma_{\text{inc}}/\Gamma^* \leq \exp(\text{UB} - \text{LB})$.

Second, we prove that semantically equivalent circuit representations can alter the optimal fragment placement itself. We establish an upper bound on representation regret, prove that a perturbation $\|\delta\|_1$ smaller than the primary optimality margin m_a guarantees optimal-partition containment ($\arg \min J_b \subseteq \arg \min J_a$), and prove a positive-scaling invariance corollary explaining why QAOA placement is invariant ($R = 1$) despite a $2\times$ cost shift. Using a tie-safe two-stage lexicographic MIP across multiple solver tolerances, we confirm placement reversals on QPE ($R = 729\times$) and Grover circuits ($\log_{10} R = 62.8$), while ruling out single-solve tie-breaking artifacts on QFT circuits.

The solver evaluations show that basic SCIP closes 200/200 heterogeneous $K = 2$ E7 instances within 10 s and 101/144 K -way E9 runs through $n = 60$, with explicit solver-tolerance factors reported for all open cases. CERTICUT provides a rigorous formulation, resource-interpretable anytime-bound semantics, and a reproducible reference implementation for independent-QPD circuit-cutting optimization.

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APPENDIX A

FAMILY-LEVEL RESULTS FOR LEGACY E1 B2S

REFERENCE AT $n = 40$

Table XII breaks down the largest size in the legacy E1 corpus ($n = 40$) by graph family. Median time ranges from 0.809 s for community instances to 37.215 s for weighted-random instances. These values describe the evaluated samples; they do not establish a causal relationship between graph structure and computational difficulty.

TABLE XII
FAMILY-LEVEL PERFORMANCE BREAKDOWN AT $n = 40$. EACH FAMILY
CONTAINS TEN SEEDED INSTANCES.

Family	Solver-tol. closed	Med. time (s)	p90 time (s)
Community	10/10	0.809	1.186
Nearest-neighbor	10/10	1.646	1.764
QAOA ring	10/10	5.367	6.173
Noisy-community	9/10	2.419	18.188
Random	10/10	18.344	41.224
Dense	10/10	24.070	52.669
Weighted-random	8/10	37.215	60.067

APPENDIX B BOUND-REPORT SEMANTICS

We evaluate F in the log domain to avoid overflow. “Closed” means a solver-reported gap $UB - LB \leq 10^{-9}$; it is not an independently verified exact proof. “Bound report unavailable” means that no safe boundary has completed by the checkpoint. At 60 s, all E1 runs have finite bounds; the three open runs retain positive gaps and nonempty frontiers and are classified as time-limit bound reports.

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